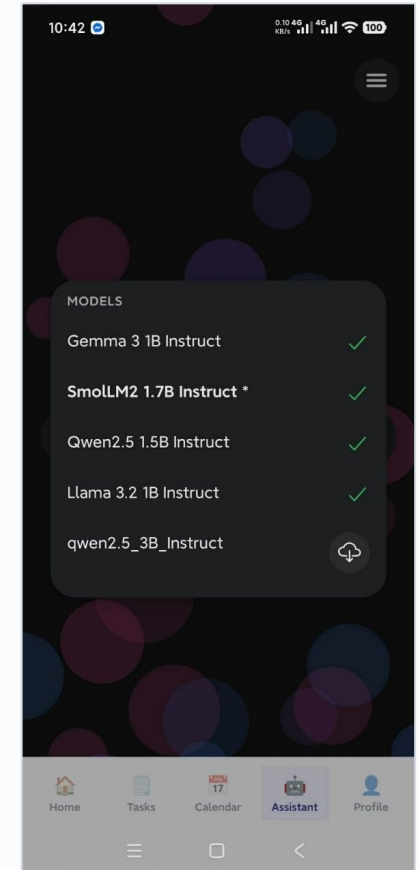
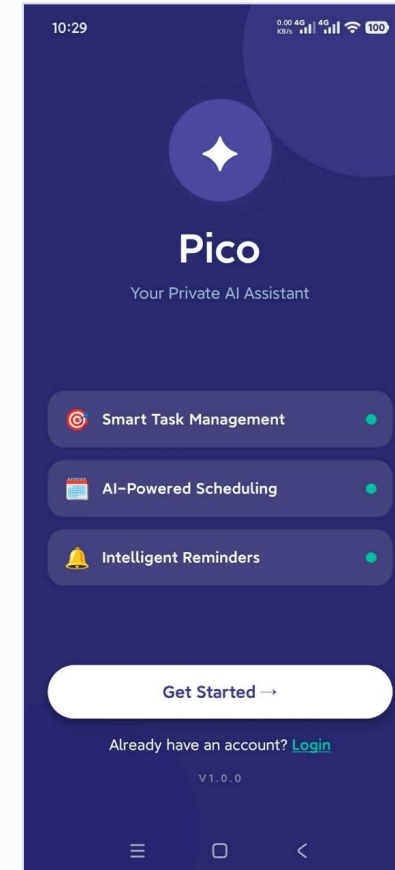


USER MANUAL

Pico

A private AI assistant that runs on your phone. The model lives on the device, so Pico manages your tasks, schedule and hardware without sending a word of it anywhere.

Version 1.0.0 · Android · CSE-464



What Pico does

Six capabilities that share one brain, rather than six features side by side.

On-device AI

A small language model runs locally through llama.cpp. Five models to choose from; no account, no server, no network call.

Task management

Create, search, filter, complete and delete tasks with a priority and a due date. Three ways in: the form, the tool menu, or plain chat.

Calendar

A month grid with event density dots and a day agenda, merging your device calendar with the events Pico creates itself.

Device tools

Sixteen actions Pico can take on the phone: flashlight, battery, location, timers, weather, maps, search and notifications.

Telegram sync

Point Pico at a Telegram bot and it turns incoming notices into tasks, new events, or reschedules of events you already have.

Private by design

Tasks, events, reminders and accounts sit in encrypted on-device storage. Nothing is uploaded, because there is nowhere to upload it.

What joins them together

Every one of these is a tool behind a single contract, listed in one registry. The chat, the tool menu and the notification buttons all call the same tools by the same names - which is why anything Pico can do by button, it can also do by sentence.

How a request travels

Pico answers at whichever tier can answer first, so the model only runs when it is actually needed.



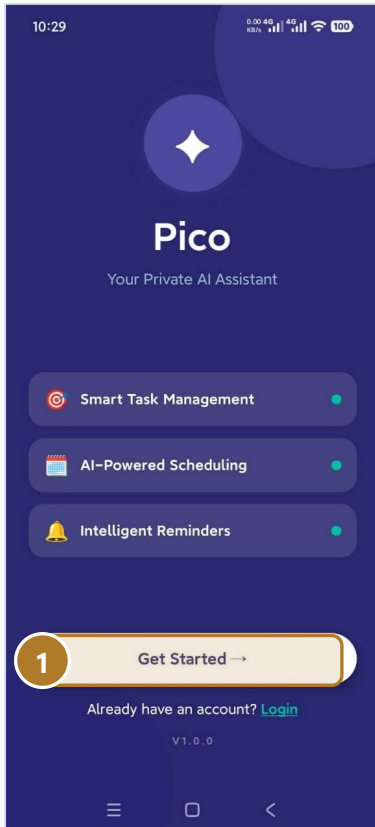
Tier	When it runs	Speed	Needs the model?
Fast path	The phrase matches one of 15 intent rules - greetings, flashlight, battery, tasks, weather, timers, search	Instant	No
On-device model	Anything else: free conversation, or an action phrased in a way the rules did not anticipate	1-10 s on CPU	Yes - must be downloaded
Cloud escalation	Plan My Day and Telegram sync only, and only when a key was built in; silently falls back to on-device	1-3 s	No - falls back to it

Why it is built this way

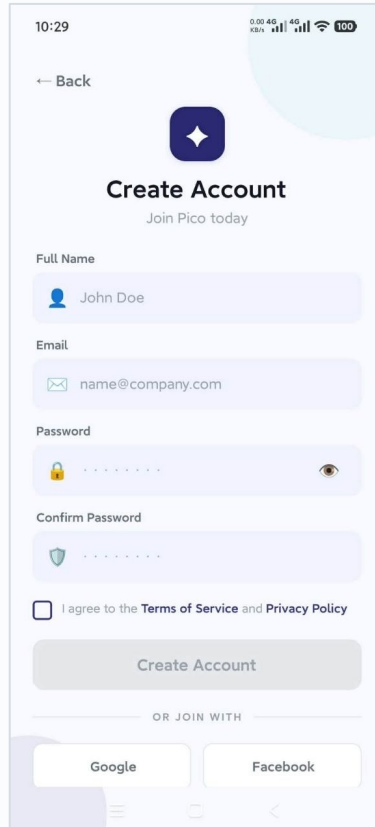
A 1-billion-parameter model on a phone CPU is slow and occasionally wrong. Routing the obvious commands around it makes the common case instant and leaves the model for what only it can do.

Create an account

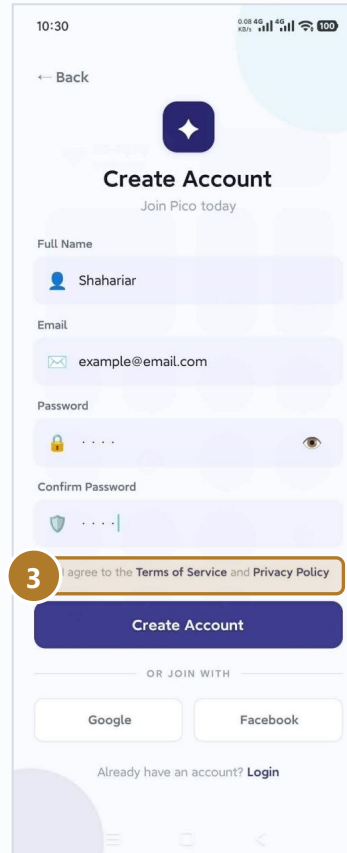
There is no server and no verification email - the account is made and checked entirely on the device.



Splash



Empty form



Ready to submit

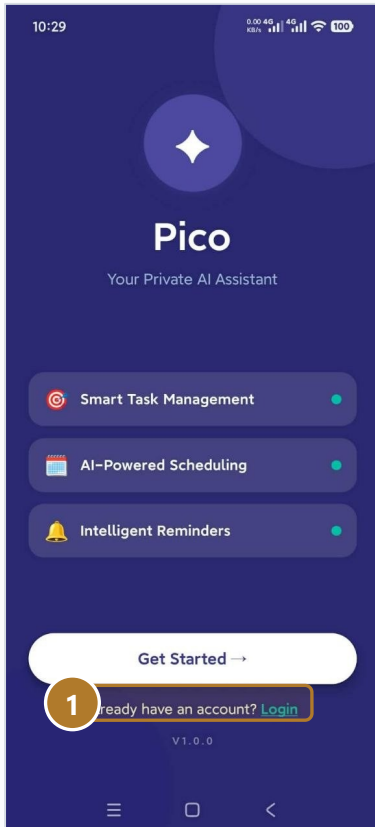
- 1 Tap Get Started on the splash screen**
Already registered on this phone? Use the Login link underneath it instead.
- 2 Fill in your name, email and password**
The password needs at least 6 characters, and Confirm Password has to match it exactly.
- 3 Tick the Terms of Service box**
Create Account stays greyed out until you do - sign-up is refused without consent.
- 4 Tap Create Account**
No confirmation code arrives, because there is no mail server. You are signed straight in and land on Home.

The Google and Facebook buttons do nothing

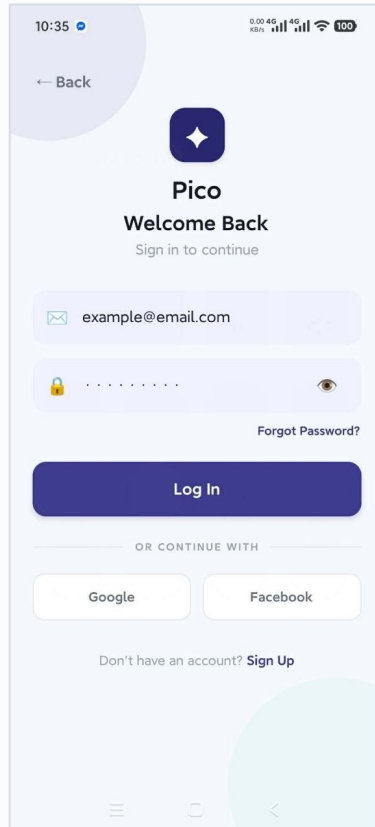
They are leftover UI from the design mock-up, as is Forgot Password. Use email and password.

Log in

Only needed after you sign out - otherwise Pico keeps you signed in between launches.



Splash



Welcome Back

- 1 From the splash screen, tap Login**
Underneath Get Started, on the "Already have an account?" line.
- 2 Enter the email you registered with**
Matching ignores capitalisation, so the exact case does not matter.
- 3 Type your password - the eye reveals it**
Useful for checking a long password before submitting it.
- 4 Tap Log In**
A wrong email or password raises "Invalid email or password". Since accounts live only on this phone, one created on another device will not be found here.
- 5 You stay signed in from then on**
The session is held in encrypted storage and survives restarts. Only Profile > Sign Out ends it.

Switch to dark mode

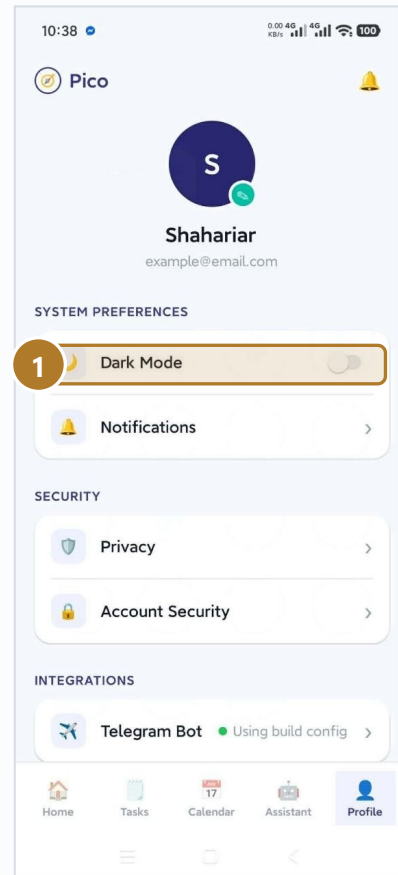
One switch in Profile, applied across the app the moment you flip it.

Where

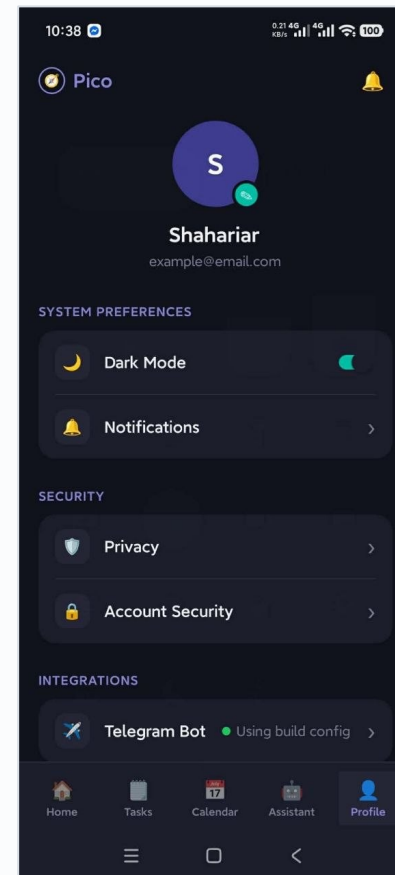
Profile tab > System Preferences > Dark Mode.

Scope

Home, Tasks, Calendar and Profile all change at once. The Assistant is always dark by design, and the sign-in screens are always light.



Light



Dark

Default

Light. Pico does not follow the phone's own light or dark setting.

Not remembered

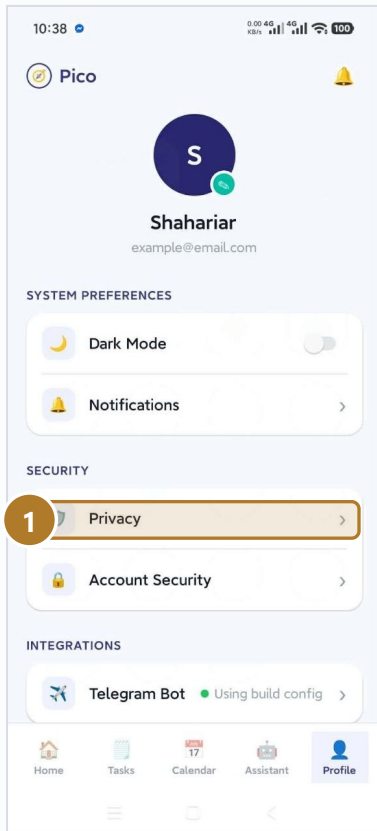
The choice is held in memory only, so closing the app returns it to light. Worth knowing before you demonstrate it.

Worth showing

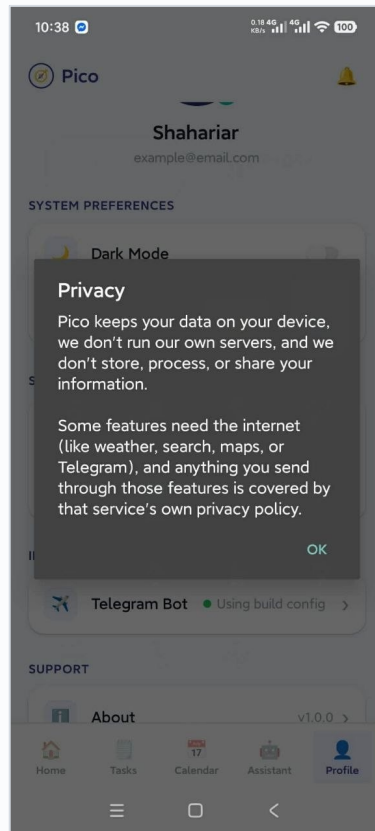
A two-second flip that proves the whole theme system works.

Read the privacy notice

Pico states plainly what stays on the phone and what does not.



Profile > Security



The notice

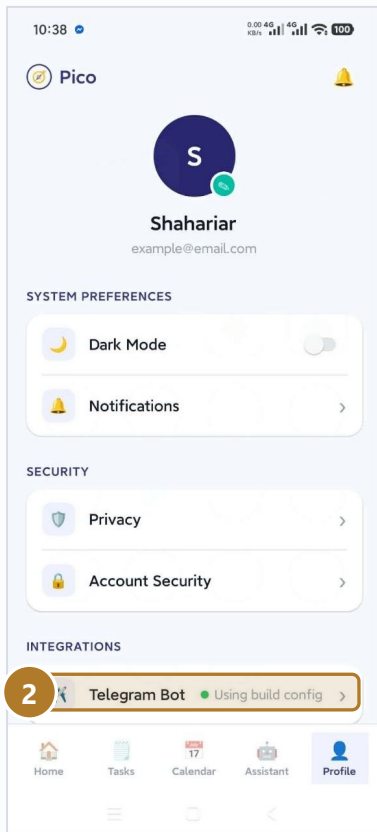
- 1 Profile tab > Security > Privacy**
The shield row, directly above Account Security.
- 2 Read what Pico does and does not do**
Your data stays on the device; Pico runs no servers of its own and does not store, process or share your information.
- 3 Note the exception it names**
Weather, search, maps and Telegram need the internet, and anything sent through them is covered by that service's own privacy policy.
- 4 Tap OK to close**
Nothing is recorded and no consent is collected - it is a statement, not a prompt.

Why this claim holds

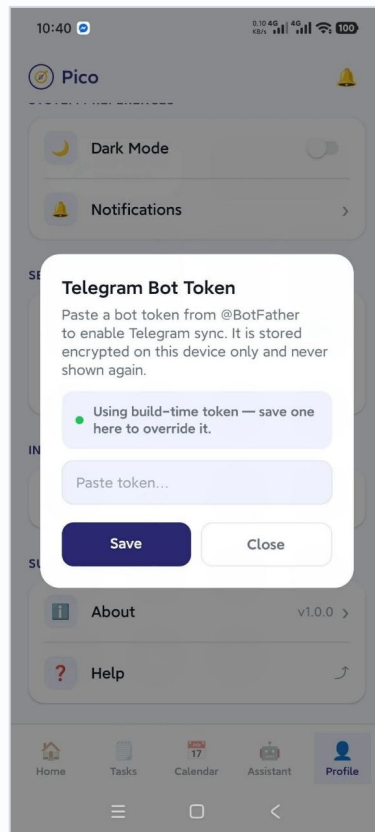
There is no Pico backend at all. Tasks, events, reminders and accounts are written to the phone's encrypted Keystore-backed storage, and the AI model runs on the device's own CPU - so your prompts are never transmitted to be answered.

Set up the Telegram bot key

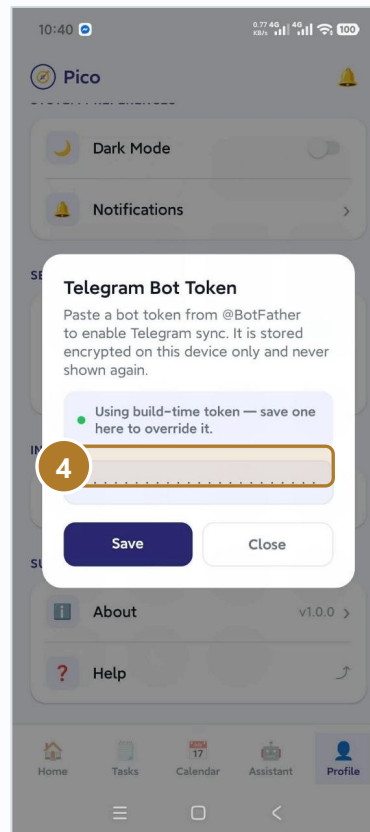
The setup step behind the Telegram updates tool - without a token, that tool only reports that one is missing.



Integrations



Token dialog

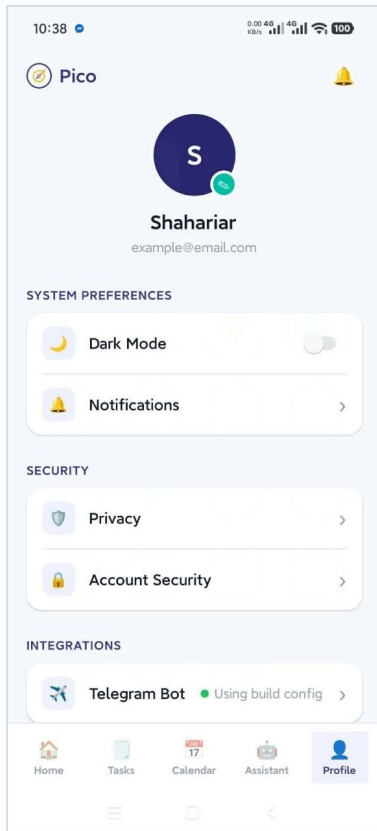


Pasted

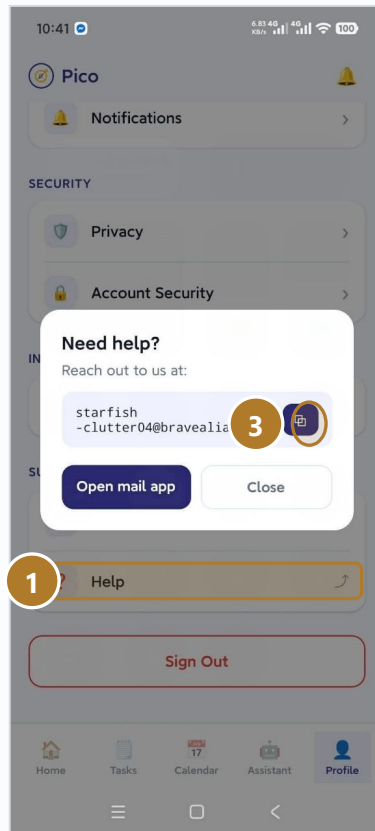
- 1 Create a bot in Telegram with @BotFather**
It replies with a token. Send your new bot a message or two so the sync has something to read.
- 2 Profile tab > Integrations > Telegram Bot**
The row shows its current state: Connected, Using build config, or Not connected.
- 3 Paste the token into the dialog**
It is written to encrypted device storage and never displayed again - the field shows dots from then on.
- 4 Tap Save**
A saved token overrides any build-time one. Remove saved token also resets the sync position, so a new token starts clean.
- 5 Run a sync to confirm**
Assistant > tool menu > Telegram updates, or type "check telegram". Only messages newer than the last sync are read.

Reach out for help

One dialog with the support address, a copy action and a direct route into your mail app.



Profile

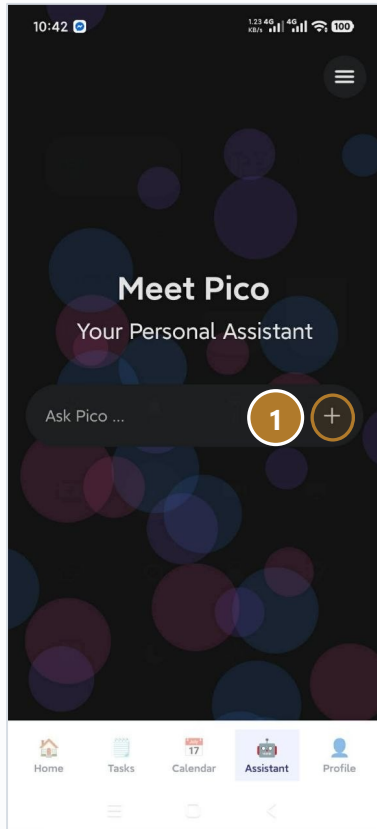


Need help?

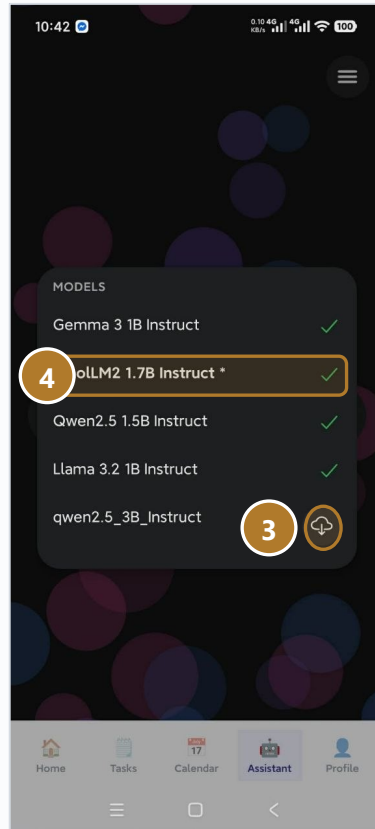
- 1 Profile tab > Support > Help**
The question-mark row, under About.
- 2 The support address is shown and selectable**
You can long-press to select the text directly if you would rather not use either button.
- 3 Tap the copy icon to share the address**
It opens the phone's share sheet with the address as the message, so you can send it to yourself or paste it elsewhere.
- 4 Or tap Open mail app**
This composes a new email to support with the subject line "Pico Help" already filled in.
- 5 Close returns you to Profile**
About, on the row above, opens the project's GitHub page in your browser instead.

Download an on-device model

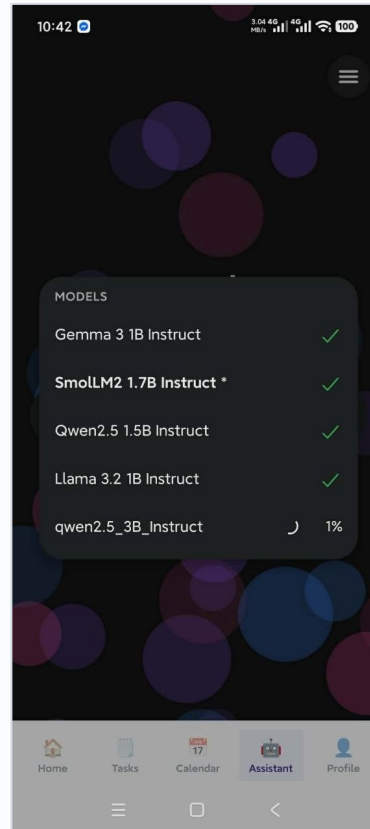
Until a model is on the phone, Pico answers only from the fast path. This is the one step that needs the internet.



Assistant tab



The model list

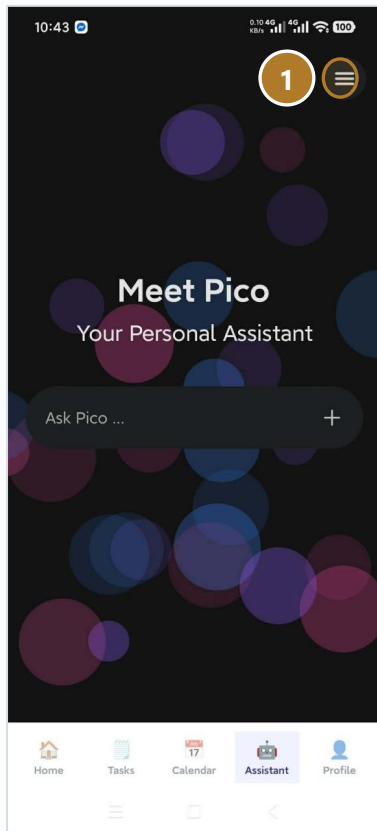


Downloading

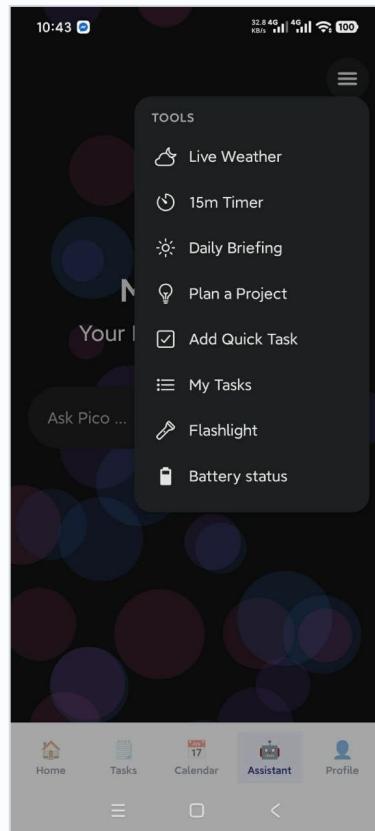
- 1 Open the Assistant tab and tap +**
The plus beside the chat box opens the model sheet. It is the only way in - there is no model setting in Profile.
- 2 Pick from the five models offered**
Gemma 3 1B Instruct, SmolLM2 1.7B, Qwen2.5 1.5B, Llama 3.2 1B and Qwen2.5 3B. Smaller models answer faster.
- 3 Tap the cloud icon and stay on Wi-Fi**
Models run from roughly 800 MB upward. Only one downloads at a time - the other rows disable until it finishes.
- 4 A tick means it is on the device**
Downloading does not select it. Tap the model's name to activate it - an asterisk marks whichever model is live.
- 5 Watch the percentage while it runs**
Leaving the sheet does not cancel the download. Switching models later keeps both on the phone.

The tool menu

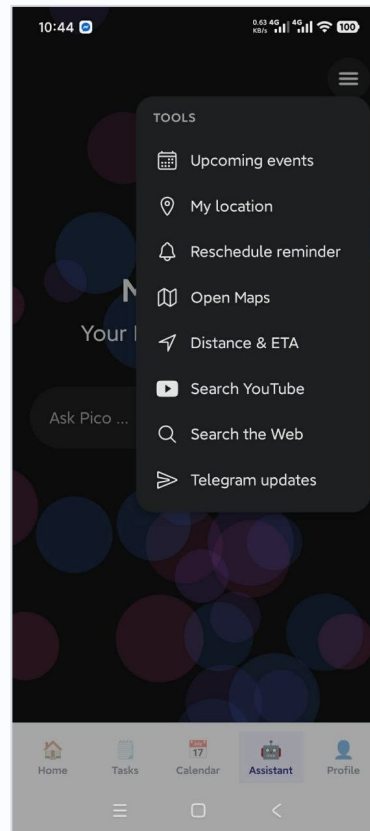
Everything Pico can do to the phone, on a button - for when you want the action without composing a sentence.



Assistant tab



Tools, top



Tools, scrolled

- 1 Tap the menu icon, top right**
It sits above the chat, clear of the message list.
- 2 Eight actions are visible at first**
Live Weather, 15m Timer, Daily Briefing, Plan a Project, Add Quick Task, My Tasks, Flashlight and Battery status.
- 3 Scroll for the other eight**
Upcoming events, My location, Reschedule reminder, Open Maps, Distance & ETA, Search YouTube, Search the Web and Telegram updates.
- 4 Tap one and it runs immediately**
A spinner shows on that row while it works, then the menu closes and the result is posted into the chat as Pico's reply.
- 5 The model is never consulted here**
Buttons call the tools directly, so every one of them works even with no model downloaded.

What each of the sixteen tools does

The same tools the model calls when you phrase a request in chat.

Tool	What it does	Needs
Live Weather	Temperature, feels-like, high and low, humidity, wind, rain chance	Internet + location
15m Timer	A 15-minute focus timer that rings as a notification	Notifications
Daily Briefing	Battery, weather, today's events and pending tasks in one card	Offline
Plan a Project	Breaks a goal into 3-4 subtasks and saves them to your list	Model or cloud
Add Quick Task	Writes one fixed demo task at High priority	Offline
My Tasks	Lists your pending tasks, five at a time	Offline
Flashlight	Toggles the torch, reading its current state first	Camera
Battery status	Charge level and whether the phone is charging	Offline

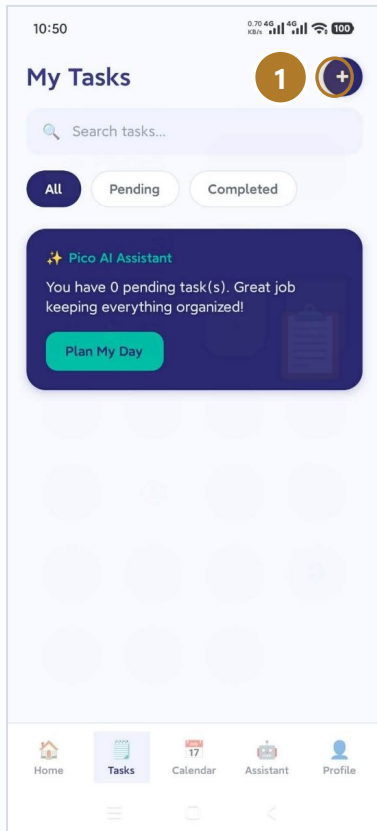
Tool	What it does	Needs
Upcoming events	Device calendar events for the next seven days	Calendar
My location	Current GPS latitude and longitude	Location
Reschedule reminder	Fires a notification carrying Yes / No buttons	Notifications
Open Maps	Hands off to the Google Maps app	Internet
Distance & ETA	Road distance and drive time, without live traffic	Internet + location
Search YouTube	Opens YouTube search results for a query	Internet
Search the Web	Opens a Google search in the browser	Internet
Telegram updates	Fetches bot notices and syncs them to tasks and events	Internet + token

Four of these are button-only

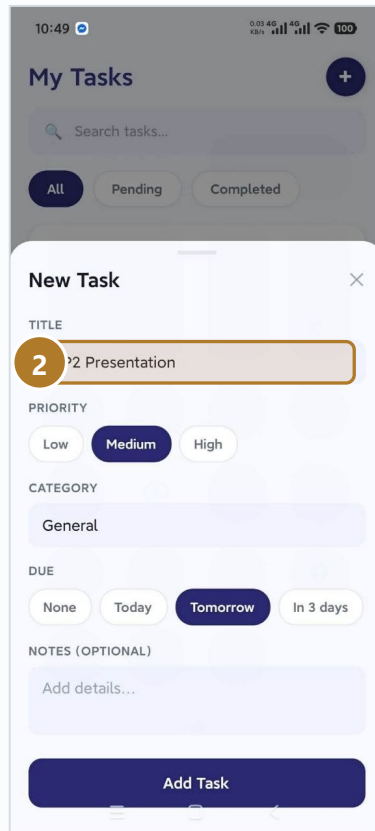
Distance & ETA, Open Maps, Search YouTube and Search the Web are not in the list the model is shown, so it cannot call them on its own. Reach them from this menu, or with the exact chat phrasings in the cheat-sheet - "how far is Gulshan 2", "navigate to...", "play...", "search for...".

Add a task

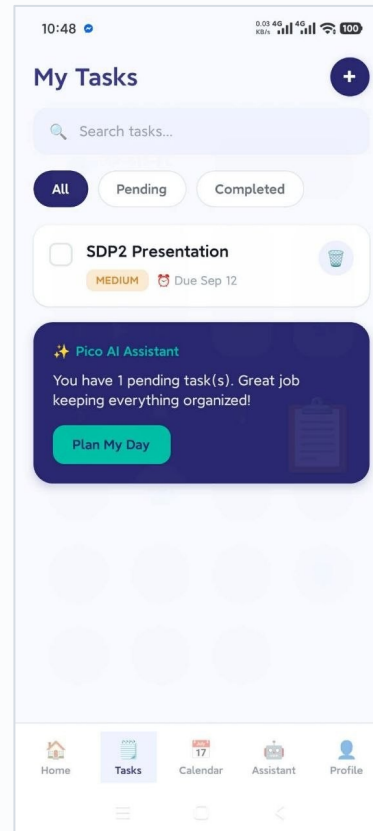
The form gives you every field. Chat and the tool menu reach the same list by other routes.



Empty list



New Task

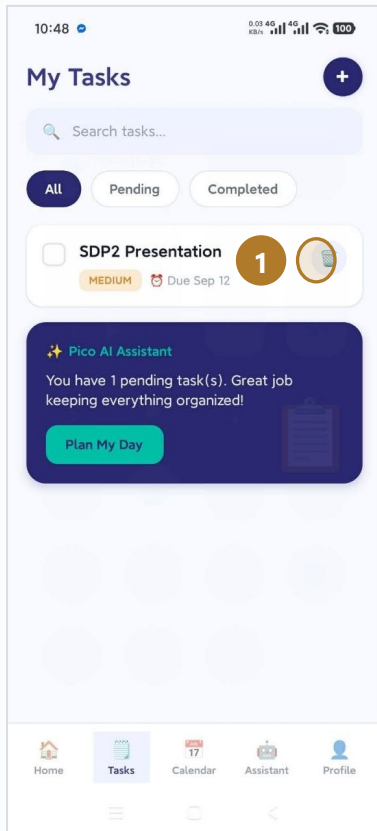


Saved

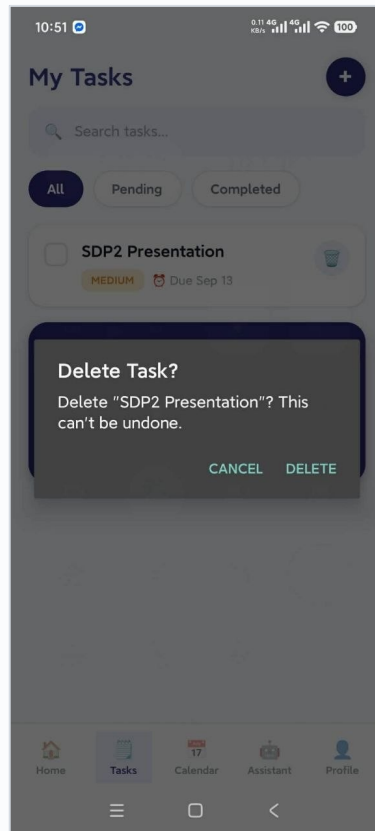
- 1 Tasks tab > the + button**
Top right. A New Task sheet slides up from the bottom.
- 2 Give it a title**
The only required field. Saving without one raises "Add a title" and nothing is written.
- 3 Choose a priority**
Low, Medium or High, defaulting to Medium. This drives the colour of the pill in the list and the High Priority count on Home.
- 4 Set a category**
Free text, defaulting to General. Tasks created by the Telegram sync arrive tagged Telegram.
- 5 Pick a due date, and add notes if you want**
None, Today, Tomorrow, In 3 days or In 1 week - each resolved to midnight on that day. Notes are optional.
- 6 Tap Add Task**
It appears in the list immediately, and every counter in the app updates with it.

Delete a task

Deliberately two steps, because there is no undo.



The task list



Confirm

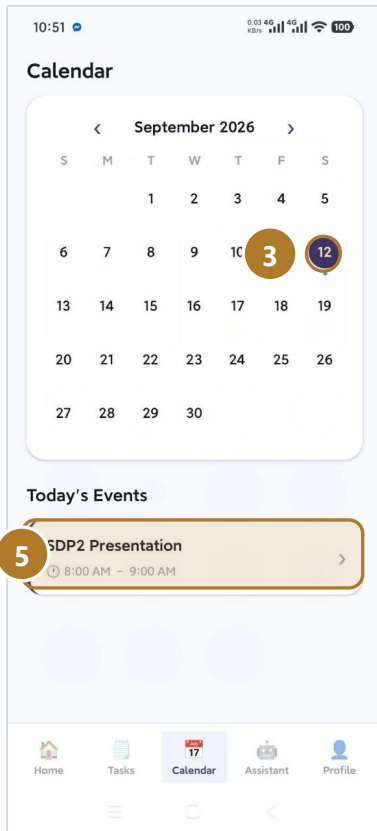
- 1 Tap the bin icon on the task**
On the right-hand edge of the card. It has a generous tap target, so it is hard to miss and hard to hit by accident.
- 2 Read the confirmation - it names the task**
"Delete 'SDP2 Presentation'? This can't be undone." Naming the task means you cannot delete the wrong one by muscle memory.
- 3 Tap DELETE to remove it, CANCEL to back out**
Deletion is permanent - there is no bin to recover it from, and the data export will not contain it afterwards.
- 4 To keep a task but mark it done, use the checkbox instead**
Completed tasks stay in the list with a strikethrough and show under the Completed filter.

Deleting everything at once

Profile > Account Security > Delete all data erases every task, event and reminder together, along with your account. Step 14 covers it.

View the calendar

A month at a glance, then the agenda for whichever day you tap.



Calendar

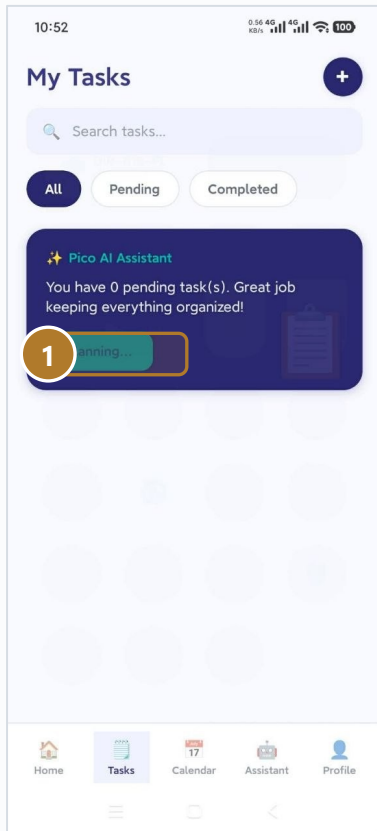
- 1 Open the Calendar tab**
It asks for calendar permission the first time. Refuse it and the screen shows a lock with a Grant access button instead of a grid.
- 2 Move between months with the arrows**
Jumping to another month selects day 1; coming back to this month re-selects today.
- 3 Today is ringed, the selected day is filled**
Two different states, so you never lose track of where today is while browsing ahead.
- 4 Dots under a day count its events**
Up to three, then it stops - the dots show how busy a day is, not an exact number.
- 5 Tap a day to read its agenda**
Listed under the grid, all-day events first and then by start time, each with its calendar's own colour down the left edge.
- 6 Tap an event to open it in the phone's calendar**
Pico hands off to the native calendar app at that date. Pull down on the list to refresh.

Two calendars feed this screen

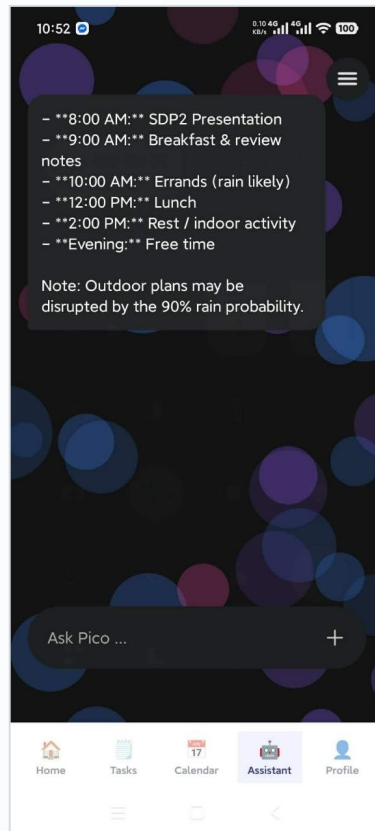
Your real Android calendar, and the separate store Pico writes to when it creates an event from chat. This screen merges both - but an event Pico made will not appear in Google Calendar.

A tool call end to end: Plan My Day

The clearest demonstration of the pipeline - one tap runs several tools and hands their output to the model.



Planning...



The plan, in chat

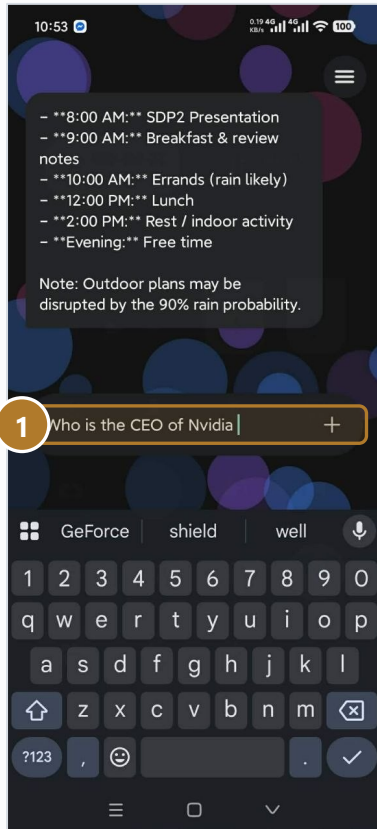
- 1 Tap Plan My Day**
On the Pico AI Assistant card, at the bottom of Tasks or on Home. Typing "plan my day" in the chat does exactly the same thing.
- 2 The button reads Planning... while it works**
Behind it, a planner decides which read-only tools to run - normally your tasks, your calendar and the weather.
- 3 Those tools run and their output is collected**
A tool that cannot reach the internet is marked unavailable rather than failing the whole plan.
- 4 The finished plan is posted into the Assistant**
An hour-by-hour schedule built from what you actually have on, plus a note about anything that might disrupt it - here, a 90% rain chance against outdoor plans.

Where the writing happens

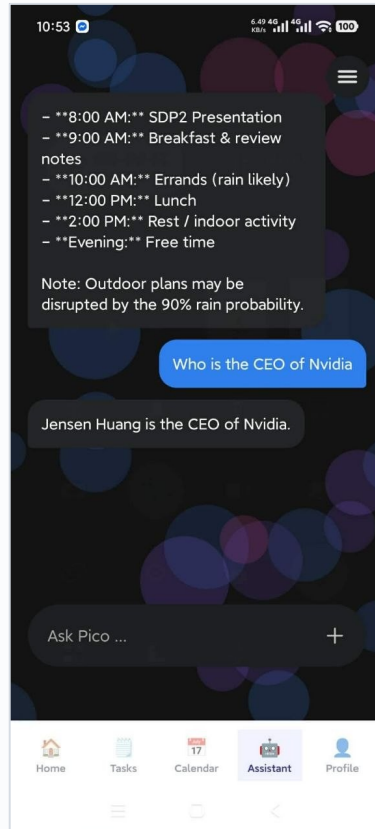
Plan My Day tries the cloud model first and falls back to the on-device one when there is no key or no signal. If both are unreachable it still answers, using the deterministic daily briefing instead - so the button never simply fails.

Just talking to Pico

When nothing needs doing, the on-device model answers in plain prose.



Typing

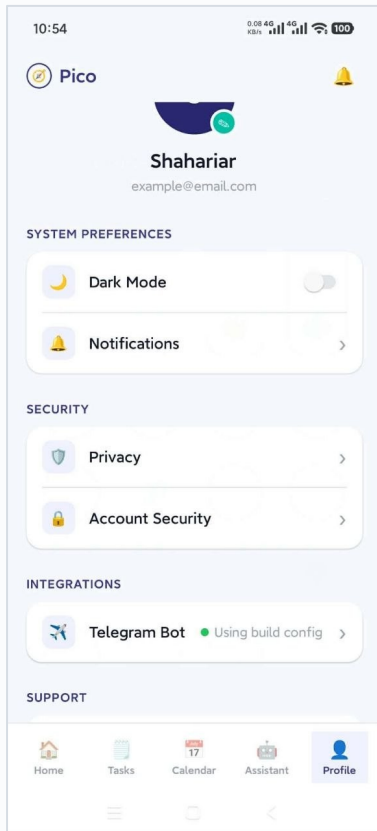


The answer

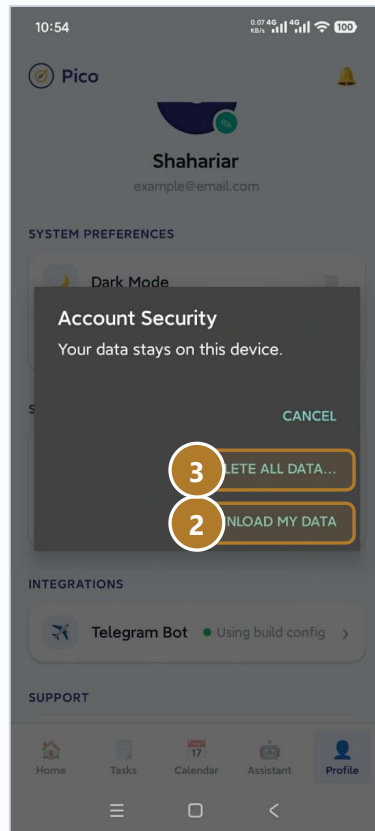
- 1 Type into Ask Pico and send**
The send key on the keyboard submits. Input locks while Pico is working, so a question cannot be sent twice.
- 2 Pico is thinking means the model is running**
It only appears on this path. Fast-path commands answer with no indicator at all, because there is no wait.
- 3 Your message is the blue bubble on the right**
Pico's replies are the dark bubbles on the left.
- 4 The answer comes from the model on the phone**
Nothing is looked up and nothing is transmitted - which also means answers are limited to what a 1B-parameter model knows, and it can be confidently wrong.
- 5 For anything live, ask it to search**
"search for..." opens a real web search in the browser instead of guessing from the model's memory.

Download or delete your data

Both controls sit behind one row, and one of them is irreversible.



Profile > Security



Account Security

- 1 Profile tab > Security > Account Security**
The padlock row. The dialog opens with a reminder that your data stays on this device.
- 2 Download my data exports everything as JSON**
Your tasks, events and reminders, with a timestamp, handed to the phone's share sheet so you can mail it to yourself or save it.
- 3 Delete all data erases the lot**
Tasks, events, reminders, the saved Telegram token, the Telegram sync position, every account on the device - and it signs you out.
- 4 There is no undo and no grace period**
The erase happens the moment you confirm. Export first if you want a copy.

What the export never contains

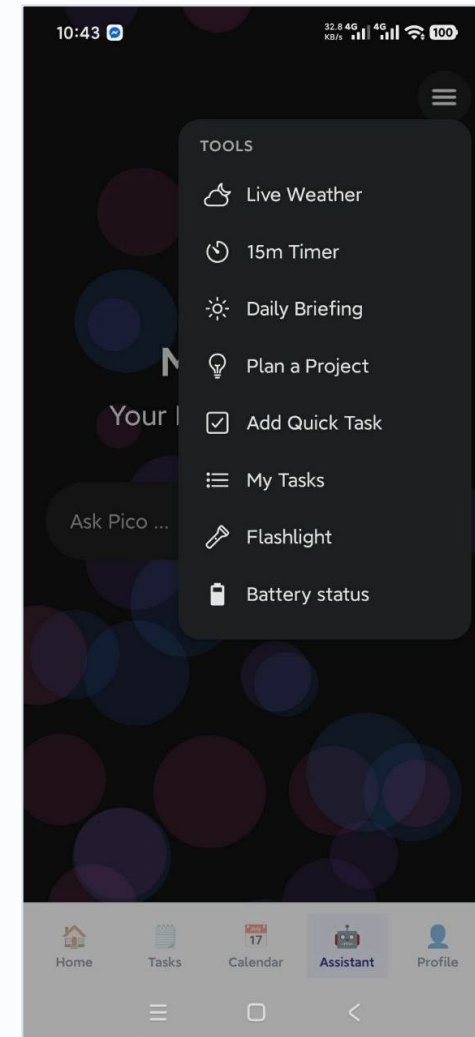
Passwords are not included. They are held separately, and the export is written from your tasks, events and reminders only.

R

Reference

Where each feature lives, everything you can say to the Assistant, and what to do when something misbehaves.

- A feature-to-screen lookup table
- The complete Assistant command cheat-sheet
- Offline and online, feature by feature
- Troubleshooting for what actually goes wrong



Tool menu

Where everything lives

Every feature in Pico and the shortest route to it.

I want to...	Go to	Notes
Create an account	Splash > Get Started	Local only - no email verification
Log back in	Splash > Login	Only needed after signing out
Switch to dark mode	Profile > Dark Mode	Resets to light when the app restarts
Read the privacy notice	Profile > Privacy	States what leaves the device
Connect Telegram	Profile > Telegram Bot	Needs a token from @BotFather
Contact support	Profile > Help	Copies the address or opens your mail app
Download or change the AI model	Assistant > +	Tap a name to activate it; * marks the live one
Run a device action by button	Assistant > menu icon	Sixteen tools, no typing
Add a task with full detail	Tasks > +	The only route that accepts notes
Delete a task	Tasks > bin icon	Confirms by name; cannot be undone
See my month	Calendar tab	Dots show density; tap a day for its agenda
Plan my day with AI	Tasks or Home > Plan My Day	Result is posted into the Assistant chat
Just ask a question	Assistant > Ask Pico	Needs a downloaded model
Export or erase everything	Profile > Account Security	JSON export, or an irreversible wipe

Assistant cheat-sheet: instant commands

Matched before the model is consulted, so they answer with no wait - and work even with no model downloaded.

Say this	And Pico will	Also matches
hi / hello / hey	Greet you	hola, sup
who are you / what can you do	Introduce itself and list its abilities	-
what is today / what time is it	Give the full date and the clock time	today's date, current date
turn on the flashlight	Switch the torch on	torch on, light on
turn off the flashlight	Switch it off	torch off, kill the light
battery	Report charge and charging state	how much juice, power level
my tasks	List pending tasks, five at a time	show todos, what tasks do I have
upcoming events	Read the next seven days of calendar	my calendar, my schedule, agenda
where am I	Give GPS coordinates	my location, current location
daily briefing	Battery, weather, today's events, pending tasks	brief me, good morning, how is my day
mark buy groceries as done	Complete the matching task	complete X, finish task X
plan my day	Build a plan from tasks, events and weather	plan today, what should I do today
plan my final presentation	Break the goal into 3-4 saved subtasks	break down X, help me prepare for X
timer for 20 minutes	Schedule a notification	set an alarm, countdown
weather	Full local forecast	forecast, do I need an umbrella

Cheat-sheet: the rest

The remaining instant commands, then everything that hands over to the on-device model.

STILL INSTANT

Say this	And Pico will
how far is Gulshan 2	Road distance and drive time
navigate to Dhaka University	Hand off to Google Maps
play Bohemian Rhapsody	Open YouTube search results
search for the UCL score	Open a Google search
check telegram	Fetch and sync bot notices

NEEDS THE MODEL - EXPECT A SHORT WAIT

Say this	And Pico will
add a task to buy milk tomorrow, high priority	Create it, reading title, priority and category out of the sentence
schedule a meeting on Friday at 3pm	Create an event, parsing the date and the time
who is the CEO of Nvidia	Answer from the model's own knowledge, offline
anything conversational	Reply in prose rather than running a tool

If a command is not recognised

Phrase it the way the cheat-sheet does. The instant tier matches on keywords, so "illuminate the room" will not find the flashlight - it goes to the model, which may or may not work it out. Four tools - distance, maps, YouTube and web search - are reachable only through these phrasings and the tool menu; the model cannot call them.

What works offline, and what does not

There is no Pico server. The features that use the internet talk to third parties directly.

Where your data lives

Tasks, events, reminders, accounts and the Telegram token sit in expo-secure-store, encrypted by the Android Keystore.

Where the AI runs

On the phone's CPU through llama.cpp. Your prompts and your task list are never sent anywhere to be answered.

What is never collected

No analytics, no telemetry, no crash reporting, no advertising ID - there is no server account to hold them.

How to get it all back

Profile > Account Security > Download my data exports everything as JSON. Delete all data erases it irreversibly.

Feature	Offline?	What it contacts
Sign in, tasks, events, reminders, calendar	Fully offline	Nothing
Chat, briefings, goal breakdown, flashlight, battery, timers	Fully offline once a model is downloaded	Nothing
Weather	Online	Open-Meteo - coordinates only, no identifier
Distance and ETA	Online	Photon geocoder, then the public OSRM router
Maps, YouTube, web search	Online	Hands off to the relevant app or site
Telegram sync	Online	api.telegram.org, carrying your bot token
Plan My Day, Telegram interpretation	Falls back to on-device	NVIDIA NIM, only if a key was built in

When something does not work

The failures that actually happen, and what each one means.

The chat says the model was not found

No model has been downloaded, or the active one was removed. Open Assistant > + and download one. Tool-menu actions keep working.

The first message after opening is slow

The model is being loaded into memory. It happens once per launch; every message after it is quicker.

Every reply is slow

The model runs on the CPU with no GPU offload. A 3B model on a mid-range phone will feel heavy - switch to Gemma 3 1B.

A model download will not start

Only one downloads at a time; the other rows stay disabled until it finishes. Check free space - models run from 800 MB upward.

The calendar is empty or shows a lock

Permission was refused. Tap Grant access; after a second refusal Android stops asking and Pico opens system settings instead.

Login says invalid email or password

Accounts live only on the phone that created them. An account made on another device will not be found here.

Dark mode was light again after a restart

Expected - the theme is held in memory only and is not saved between launches.

Telegram says the token is missing

No bot token is saved: Profile > Telegram Bot. If a sync returns nothing, those messages were already read - the offset only moves forward.

An event created in chat is missing

It is not lost. Events Pico creates go to its own encrypted store, which the Calendar tab merges into the view - but the Notifications list and the Home bell badge read the Android calendar only, so it is not counted there.

Thank you

Questions welcome - on any workflow in this manual, or on what runs behind it.

[Assistant](#)[Tasks](#)[Calendar](#)[Telegram](#)[On-device AI](#)[Privacy](#)